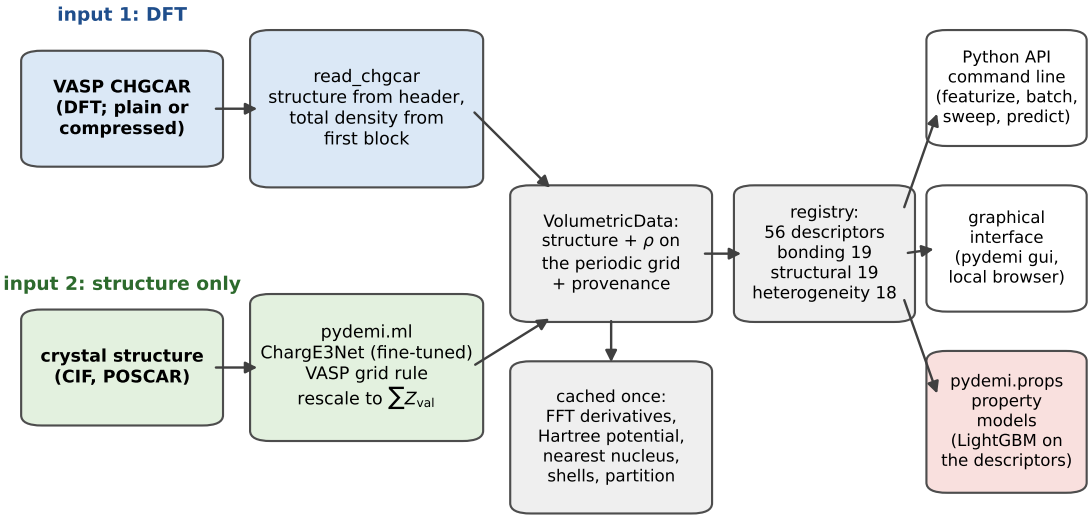
input 1: DFT
VASP CHGCAR (DFT; plain or compressed)
read_chgcar structure from header, total density from first block
input 2: structure only
crystal structure (CIF, POSCAR)
pydemi.ml ChargE3Net (fine-tuned) VASP grid rule rescale to $\sum Z_{val}$
VolumetricData: structure + ρ on the periodic grid + provenance
cached once: FFT derivatives, Hartree potential, nearest nucleus, shells, partition
registry: 56 descriptors bonding 19 structural 19 heterogeneity 18
Python API command line (featurize, batch, sweep, predict)
graphical interface (pydemi gui, local browser)
pydemi.props property models (LightGBM on the descriptors)